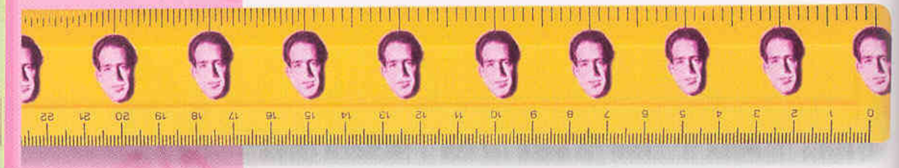

affecting reality
Niels Bohr said that what we choose to measure, and the act of measuring, affect the nature of reality. The most important feature of the Copenhagen Interpretation is that no quantum property (such as the position of an electron) is real until it is measured.

dice." All the evidence is that Einstein was wrong and the quantum world really is ruled by probability.

The conclusions drawn from the results of experiments such as the double-slit experiments were developed at the end of the 1920s. They are collectively known as the Copenhagen Interpretation because Niels Bohr and his colleagues in Denmark were closely involved in its development (although key ideas about probability came from Max Born in Germany).

The Interpretation says that a quantum system does not exist in a definite state until it is measured. For example, an electron traveling through the double-slit experiment does so as a spread-out wave, and does not have a precise location in space. It is only when it arrives at the detector screen that it makes a "choice" from the probabilities (like a tumbling die finally settling with three spots facing up) which causes the wave function to "collapse," as Bohr put it, onto a single point. Using this model, quantum entities travel as waves, but arrive as particles.

Schrödinger's paradox

The idea of an electron wave function proved very useful in uniting observation and theory, and enabled the Austrian Erwin Schrödinger to develop a description of the way that electrons behave in atoms in terms of waves. This is a much more complete description than Bohr's model, and enabled researchers such as the American Linus Pauling to explain all the principles of chemistry (how atoms interact to make molecules) in terms of quantum physics. This has led physicists to quip that "chemistry is now a branch of physics."

in the balance
Pushing Bohr's ideas to their absurd limits, Erwin Schrödinger imagined a cat that would be neither alive nor dead until its state of health was "measured" in some way.

In the 1920s, Austrian physicist **Erwin Schrödinger** (1887–1961) found a wave equation that describes the behavior of quantum entities such as electrons and other "particles." In 1933, he received the Nobel Prize for this work (jointly with Paul Dirac), but he was dismayed that the weirdness of quantum physics could not be explained in the familiar language of wave equations.

Schrödinger was particularly pleased that his wave function seemed to be bringing the common sense of familiar waves back into fundamental physics. This is why he was so horrified when he discovered that it was not possible to get rid of the probabilistic effects, and later said of his own theory, "I don't like it, and I wish I'd never had anything to do with it." He even dreamed up an imaginary "thought experiment," known today as Schrödinger's Cat Paradox (although it isn't really a paradox), to highlight what he saw as the absurdity of the Copenhagen Interpretation, which, taken literally, says that a cat can be both dead and alive at the same time.

Schrödinger failed in his attempt to fully explain electrons purely in terms of waves because it is no more true to say that an electron is a wave than to say that it is a particle. All we can say is that quantum entities such as electrons

> **"You believe in a God who plays dice, and I in complete law and order in a world which objectively exists, and which I, in a wildly speculative way, am trying to capture…even the great initial success of the quantum theory does not make me believe in the fundamental dice game, although I am well aware that your younger colleagues interpret this as a consequence of senility."**
>
> Albert Einstein, letter to Max Born (1926)